

RESEARCH ARTICLE

INSPIRING
PLASTICS
PROFESSIONALSPOLYMER
ENGINEERING
AND SCIENCE

WILEY

Development of lightweight and thermally insulative crosslinked polypropylene via rotomolding

Hibal Ahmad | Denis Rodrigue

Department of Chemical Engineering,
Université Laval, Quebec City, Canada

Correspondence

Denis Rodrigue, Department of Chemical
Engineering, Université Laval, Quebec
City G1V 0A6, Canada.
Email: denis.rodrigue@gch.ulaval.ca

Funding information

Association of Rotational Molders

Abstract

In this study, lightweight and thermally insulative crosslinked polypropylene (xPP) parts were successfully produced for the first time using a straightforward dicumyl peroxide (DCP) crosslinking technique via rotational molding. A novel crosslinked and cellular structure was achieved by incorporating varying DCP contents (0.5–2.5 phr) during the rotomolding process. An increase in the DCP content led to a corresponding increase in both gel content and crosslink density, while leading to lower density. In particular, the addition of 2.5 phr DCP significantly increased the cell size by 33% (528–704 μm) and cell density by 540% ($1.5\text{--}9.6 \times 10^3 \text{ cells/cm}^3$), while decreasing the density by 58% (0.902–0.378 g/cm^3). Moreover, the thermal conductivity and resistivity (insulation power) were analyzed over a temperature range (23–120°C) for different applications. The results show that the crosslinked porous structure substantially improved the thermal resistance (691%) with a very low thermal conductivity (0.055 W/m K) at room temperature. This innovative approach not only represents a significant progress in the development of lightweight and insulative materials but also sets a benchmark for future research in the field of cross-linked polymer foams, opening the door for a wide range of applications, especially for rotomolded parts.

Highlights

- The effect of DCP content on the final properties was studied.
- The incorporation of DCP into PP within the rotomolding process led to novel structures.
- Lightweight and highly insulative porous structures were obtained.
- Temperature-dependent thermal conductivity was studied for various applications.
- The crosslinked porous structure significantly enhanced thermal resistance with very low thermal conductivity (0.055 W/m K) at room temperature.

KEYWORDS

cellular structure, crosslinking, polypropylene, rotomolding, thermal insulation

This is an open access article under the terms of the [Creative Commons Attribution](#) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Polymer Engineering & Science* published by Wiley Periodicals LLC on behalf of Society of Plastics Engineers.

1 | INTRODUCTION

In recent years, the demand for insulating materials significantly increased due to a direct interest in energy efficiency and sustainability. Different industries, such as automotive, aerospace, and construction, are seeking new materials to reduce their energy consumption while providing superior thermal insulation.^{1–3} Lightweight insulative materials are especially valuable as they contribute to energy savings by reducing both the weight and energy required for transportation and thermal management.^{4–6} For example, the automotive industry benefits from lightweight materials as they reduce the overall vehicle weight, leading to improved fuel efficiency and lower emissions.⁴ In aerospace, reducing the component's weight can significantly decrease fuel consumption and increase payload capacity, enhancing operational efficiency and reducing costs.^{7,8} In construction, insulating materials play a crucial role in maintaining energy efficiency within buildings by reducing the need for heating and cooling, thus lowering energy consumption and costs.^{9,10} Additionally, acoustic insulative materials (sound barriers) are increasingly investigated in construction to improve indoor environmental quality by reducing noise pollution. This also enhances comfort and productivity in residential, commercial, and industrial buildings.¹¹

The combination of lightweight, high thermal insulation, and acoustic properties in materials is essential to meet the stringent energy efficiency standards and sustainability goals set by various industries worldwide. This drive for advanced and multifunctional engineering materials led to extensive research and development efforts on creating innovative solutions that meet these demands. To this end, polymeric foams are commonly used as insulating materials due to their excellent thermal properties, low density, and easy processing.

Polymer foams, such as polystyrene,¹² polyurethane,¹³ and polyolefins,^{14–16} have been widely adopted for various applications. They offer a combination of lightweight and effective insulation, making them ideal for packaging, building insulation, and cushioning applications.

Over the years, different technologies were developed to produce polymeric foams including batch foaming,¹⁷ bead foaming,¹⁸ foam extrusion,¹⁹ and foam injection molding.²⁰ These methods enabled the production of foams with varying densities, cell structures, and properties tailored for specific applications. Furthermore, other methods such as gas and supercritical fluid (SCF) methods offer precise control over foam characteristics by measuring solubility and diffusivity. For example, those utilizing dielectric and optical transmission techniques.^{21–23} However, monitoring the solubility and

diffusivity at high temperatures and pressures can be challenging and often requires specialized equipment and conditions.

Polypropylene (PP) is a versatile semi-crystalline polymer that has established itself as one of the most widely used plastics because of its excellent balance of mechanical properties, resistance to chemicals and moisture, as well as recyclability, making it suitable for a wide range of applications, including packaging, automotive components, and consumer goods.^{24–26} Given these advantages, PP foams hold significant potential for other areas and markets. Foamed polypropylene is especially recognized for its lightweight and exceptional durability, providing an excellent strength-to-weight ratio, making it ideal for thermal insulation.^{27,28} Additionally, PP exhibits high compatibility with other polymers, making it advantageous for the development of polymer blends in foam products. This compatibility facilitates the creation of tailored foam materials with enhanced properties by combining PP with various polymers, thereby expanding the potential applications and performance characteristics of the final foam products.²¹ This is why various studies were performed to optimize the manufacturing of PP foams, exploring different approaches such as adding nucleating agents,^{16,29,30} fibrils,^{31–33} supercritical CO₂,^{34,35} processing conditions optimization,^{36,37} and using specific additives.^{38–40} Furthermore, crosslinking was used to improve the foamability of PP using physical⁴¹ and chemical methods.⁴² In this context, crosslinking refers to the process of creating connections between polymer chains through covalent bonds, resulting in a three-dimensional network.^{43,44}

However, further research is needed to investigate simpler manufacturing techniques that do not rely on complex processes and/or numerous additives. This highlights the necessity for innovative approaches that can improve the foaming process and the overall quality of PP foams. Also, foaming within the rotomolding process is a relatively less explored area, yet it offers significant potential due to its advantages over conventional foaming techniques. Rotomolding is a four-step manufacturing process in which polymer resin is charged, heated, rotated within a mold, and finally cooled down to produce hollow and intricately shaped parts with uniform wall thickness.^{16,43,45} This technique offers several advantages over conventional foaming processes. Firstly, rotomolding excels at creating complex and large parts in a single piece, which can be challenging for conventional methods. Additionally, the rotational movement ensures a consistent wall thickness and foam density, improving the structural integrity of the final product.¹⁶

In this study, the effect of dicumyl peroxide (DCP) content (0.5–2.5 phr) on the morphological, thermal, physical, and mechanical properties of rotomolded

polypropylene (PP) is reported for the first time. A focus on the temperature-dependent thermal conductivity and resistivity over a wide range (23–120°C) is also performed. For comparison, neat PP samples were also produced. From our results, it was observed that DCP incorporation into PP parts produced via rotomolding led to an interesting discovery: the formation of a foam-like structure in crosslinked PP (xPP). This innovative material combines the benefits of crosslinking with the lightweight and insulative properties of foams, resulting in a highly insulative xPP product. Initially aimed at developing highly crosslinked PP, the unexpected porous structure shifted the focus of the study to foam development. Consequently, cellular information (cell size, density, and expansion ratio) was also studied using scanning electron microscopy (SEM) and density measurements. This foamed structure will provide potential advantages for applications where reduced weight and improved thermal insulation are required, such as in automotive interiors, thermal packaging, and insulation materials for building and construction.

The discovery of a cellular xPP via rotomolding is unprecedented in the literature and could serve as a benchmark to develop innovative materials with unique characteristics, contributing to energy efficiency and sustainability in high-performance applications.

2 | EXPERIMENTAL

2.1 | Materials

The matrix used in this study was polypropylene (RMPP141 NATURAL) from PSD Rotoworx Pty Limited (Australia). This polymer features a melt flow index of 13 g/10 min (2.16 kg/230°C) and a density of 0.902 g/cm.³ Dicumyl peroxide (DCP, bis(1-methyl-1-phenylethyl) peroxide) with 98% purity was used as the crosslinking agent and was purchased from Sigma Aldrich (USA). Irganox 1076 antioxidant (BASF, USA) was used to prevent oxidative degradation and was used as received. Xylene and toluene, purchased from Fisher Chemicals (USA), were used for Soxhlet extraction and crosslink density measurements, respectively.

2.2 | Preparation of crosslinked rotomolded parts

Selecting the appropriate temperature for rotomolding along with DCP is critical to avoid excessive degradation, as DCP has a low onset decomposition peak around 130°C, which is significantly lower than the PP melting

peak (165°C). Using DCP at higher temperatures is essential since it is most effective within a moderate range of 130–200°C. However, rotomolding typically requires elevated temperatures for optimal part formation, making careful management of these conditions crucial when using both DCP and PP. Thus, identifying the optimal heating cycle temperature and time is essential to produce high-quality xPP parts. Therefore, to determine the optimum processing conditions (with varying DCP content), trials were conducted at various oven temperatures: 172°C (DCP peak decomposition), 194°C (near endset), and above the endset (210, 220, and 230°C).⁴⁴ The cycle times were fixed at 30 min of heating and 25 min of cooling based on preliminary tests. A laboratory-scale biaxial rotomolding machine (MedKeff-Nye Roto-Lab model 22, USA) was used to produce rotomolded parts using a cubic aluminum mold with a wall thickness of 4 mm and an internal side length of 200 mm. A release agent (Trasys 420, DuPont, USA) was applied to the internal surface of the mold, and a 10 mm diameter circular vent was filled with glass wool to prevent powder loss.^{43,45} The rotational speed was set at 5:1 (arm:plate) rpm. For each test, 1 phr DCP and 0.15 phr of antioxidant (both in powder form) were mixed with the PP powder using dry-blending in a high-speed mixer (LAR-15LMB, Skyfood, USA) at 3320 rpm for 10 min. Then, an oven treatment at 85°C for 24 h was applied to achieve complete DCP penetration into the PP powder as reported in a previous work.^{43–45}

Figure 1 shows typical examples of the molded parts from these trials. It was observed that 194°C is close to the optimal condition (Figure 1B). At 172°C, the process failed to produce complete parts, as shown in Figure 1 (A), because rotomolding typically requires higher temperatures to ensure proper PP melting, flow, and sintering, which was not achieved at this low temperature. At 210, 220, and 230°C (Figure 1C–E), excessive degradation and the formation of large bubbles were observed due to the DCP having a significantly lower decomposition peak (172°C) than these temperatures. At higher temperatures, DCP decomposes too rapidly, leading to uncontrolled and localized crosslinking combined with possible PP degradation.

Based on these preliminary trials, the optimum processing conditions were determined to be an oven temperature of 194°C for 30 min and a cooling period of 25 min (forced air) with a 5:1 (arm:plate) rpm rotation. These conditions were used for the next series of experiments with varying DCP content (0.5, 1.0, 1.5, 2.0, and 2.5 phr). For comparison, neat PP parts were also produced under the same conditions, but at a higher oven temperature (270°C).¹⁶

Figure 2 presents typical images of the parts produced under these optimized conditions with varying DCP

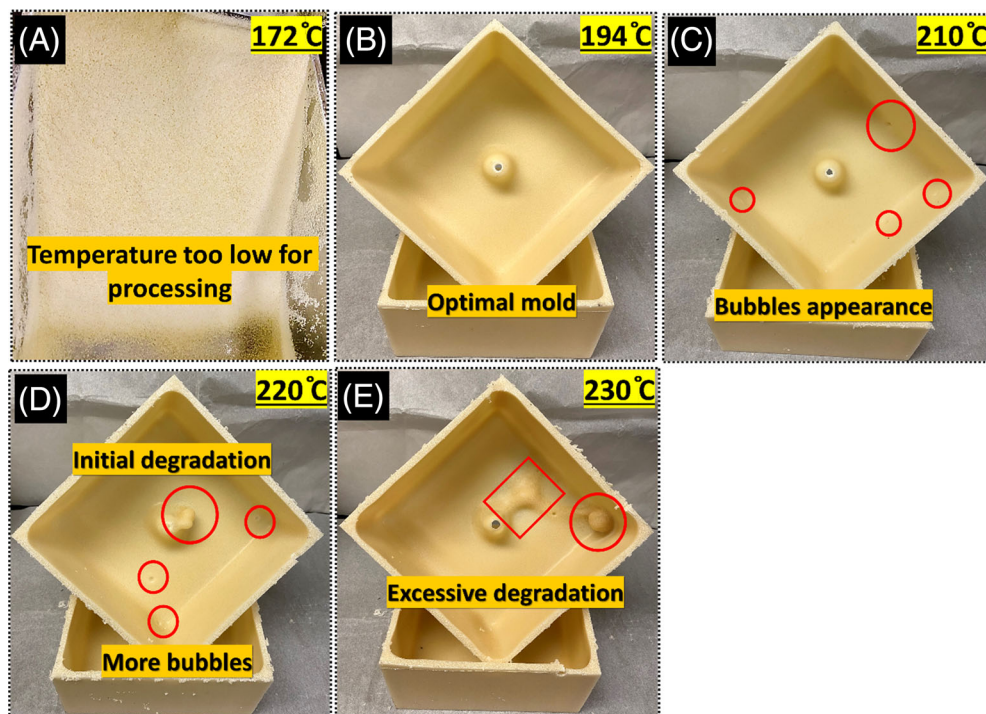


FIGURE 1 Typical examples of the rotomolded parts to determine the optimum oven temperature: (A) 172, (B) 194, (C) 210, (D) 220, and (E) 230°C.

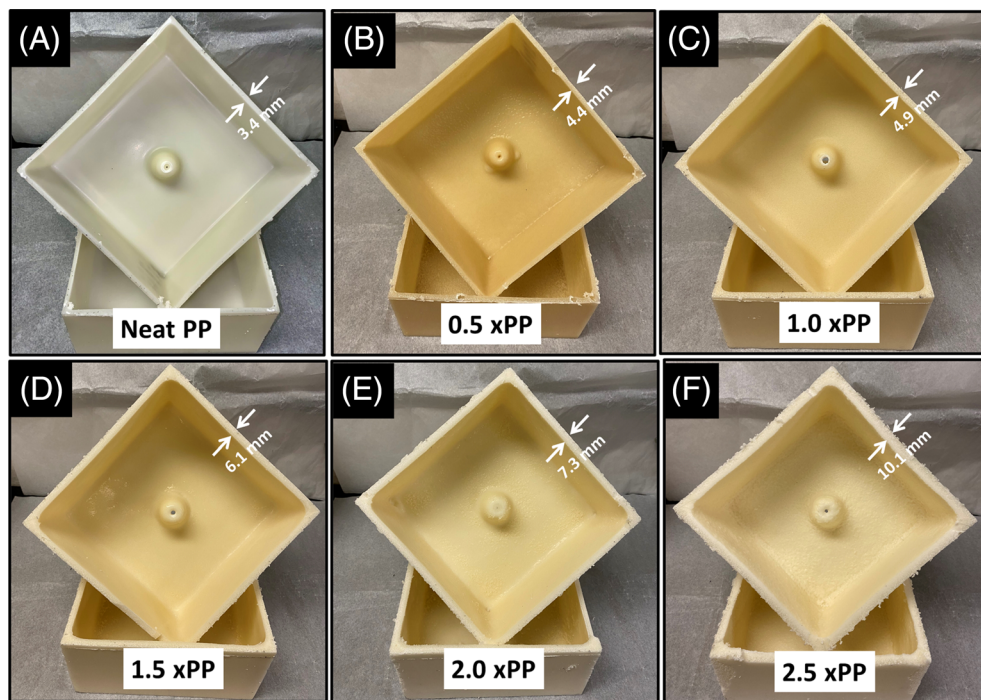


FIGURE 2 Typical rotomolded PP parts with (A) 0, (B) 0.5, (C) 1.0, (D) 1.5, (E) 2.0, and (F) 2.5 phr of DCP.

content. The images illustrate the physical differences observed with increasing DCP concentration, showcasing the effect of crosslinking on the xPP final parts (color change and increased thickness). The color change is attributed to the crosslinking process initiated by DCP, during which the decomposition of the peroxide generates free radicals. These radicals can cause slight oxidative reactions in the polymer, resulting in the formation

of colored by-products. Even with high gel content (section 3.0), the material can still take shape during the one-step crosslinking and molding process because the network formation occurs progressively as the polymer flows into the mold. This gradual crosslinking allows for sufficient mobility and shaping before the structure fully locks, maintaining some processability despite the high degree of crosslinking. These observations, along

with increased thickness, highlight the effect of DCP content on the crosslinking and structural characteristics of PP.⁴³ From the parts obtained, samples were cut for subsequent characterizations.

2.3 | Characterization

The gel content of the rotomolded xPP parts was determined by using the Soxhlet extraction method based on ASTM D2765. About 0.2–0.3 g was cut from the parts and placed in a pre-weighed stainless steel mesh pouch before being refluxed in xylene for 12 h at 140°C. Afterward, the samples were oven-dried at 85°C for 6 h until a constant weight was reached. The experiments were repeated three times to ensure accuracy, and the standard deviation was calculated to assess the experimental uncertainty.

Since polypropylene's melting point is around 165°C, it is essential to confirm that gel content measurements at 140°C provide accurate results. To ensure that the polymer was fully solubilized and validate the measurements' reliability, additional measurements (Supplementary Materials) were performed at 190°C using trichlorobenzene (TCB) for 12 h to confirm the ASTM protocol.

The gel fraction (C_{gel}) was calculated from three repetitions by determining the ratio of the insoluble polymer weight to the initial sample weight as⁴⁶

$$C_{\text{gel}}(\%) = \frac{W_f}{W_i}(100), \quad (1)$$

where W_f is the final weight after extraction and drying and W_i is the initial weight of the samples.

The crosslink density was determined by the ASTM D6814-02 equilibrium swelling method. Specimens (5–10 g) were immersed in toluene for 72 h and then weighed to determine the swollen weight (m_s). The samples were then dried overnight at 85°C and weighed again to obtain the dry weight (m_r). This process was repeated three times for each sample. The crosslink density was calculated using the Flory–Rehner equation as^{43,47}

$$v_e = \frac{-[\ln(1 - V_r) + V_r + XV_r^2]}{V_1 \left(V_r^{1/3} - \frac{1}{2}V_r \right)}, \quad (2)$$

where v_e is the crosslink density (mol/cm^3), V_1 is the solvent molar volume ($106.2 \text{ cm}^3/\text{mol}$ for toluene), X is the polymer–solvent interaction parameter (0.391), and V_r is the gel volume in the swollen sample calculated as

$$V_r = \frac{m_r/\rho_r}{m_r/\rho_r + m_s/\rho_s}, \quad (3)$$

where ρ_r is the density of the dry samples and ρ_s is the solvent density ($0.867 \text{ g}/\text{cm}^3$).

Fourier transform infrared (FTIR) analysis was performed to determine and observe the typical peaks for the neat and xPP parts. This was done by using a nitrogen-purged FTIR spectrometer (Nicolet iS50, Thermo Fisher Scientific, USA) featuring a KBr beam splitter and an MCT detector cooled by nitrogen. The spectra were recorded within the range of 3500 cm^{-1} and 500 cm^{-1} . Each spectrum was averaged over 128 scans with a resolution of 4 cm^{-1} .

Since the xPP exhibits a cellular structure, micrographs were captured at different magnifications: $75\times$ to observe the cellular structures and $1000\times$ to examine the crosslinked structure. The samples were prepared by cryogenically fracturing them in a liquid nitrogen environment. The exposed cross-sections were then vacuum coated with Au/Pd using a Leica ACE 600 (Germany). Micrographs were captured using a scanning electron microscope (SEM) on a FEI Inspect F50 (FEI, USA). The micrographs ($75\times$) were then analyzed using the ImageJ software (US National Institutes of Health, USA) to determine the cell size and cell density. The average cell size (μm) and standard deviation were calculated, while the cell density (N_f) was determined as

$$N_f = \left(\frac{n}{A} \right)^{3/2}, \quad (4)$$

where n is the number of cells in a micrograph and A is the area of the micrograph in cm^2 .

The volume expansion ratio of each sample was calculated as the ratio between the density of the original sample (ρ_s) and the measured foam density (ρ_f).

Density measurements were performed using the hydrostatic weighing method. Samples weighing 0.3–1.0 g were measured in both air and liquid (ethanol) using a precision balance (XSE 204, Mettler Toledo, USA). For samples with densities lower than that of ethanol ($0.789 \text{ g}/\text{cm}^3$), the density was calculated using Archimedes' principle, based on the mass-to-volume ratio. The average density was calculated from three samples, and the standard deviation was also determined.

The thickness of the samples was measured in millimeters (mm) using a digital Vernier caliper (Fisher, USA). Measurements were taken from the cross-sections of the samples, with the average calculated from at least 10 repetitions to ensure accuracy.

The melting and crystallization behavior was analyzed using differential scanning calorimetry (DSC 25, TA Instruments, USA). Each sample (5–10 mg) was placed in an aluminum pan, and the analysis involved heating from 20 to 180°C at a rate of 10°C/min in a nitrogen atmosphere, followed by cooling back to 20°C at the same rate. The degree of crystallinity (X_c) was calculated as

$$X_c(\%) = \frac{\Delta H_m}{\Delta H_{m0}}(100), \quad (5)$$

where ΔH_{m0} represents the melting enthalpy of 100% crystalline PP (207 J/g).⁴⁸

A homemade thermal conductivity analyzer, designed according to ASTM E1225, was used to assess the temperature-dependent thermal conductivity of xPP over a range of average temperatures (23–120°C). The rotomolded samples were cut into 50 × 50 mm² pieces, and their thickness was measured using a digital caliper (Mastercraft, Canada). The temperatures of the upper (heating) and lower (cooling) plates were set to create a 20°C temperature difference (ΔT) with average temperatures of 23, 40, 60, 80, 100, and 120°C. Each sample was placed between thin aluminum sheets to minimize thermal contact resistance. The thermal conductivity was calculated using Fourier's law as

$$k = \frac{QL}{\Delta T}, \quad (6)$$

where Q is the heat flux and L is the sample thickness. The average and standard deviation were obtained from three repetitions. More information can be obtained from a previous study.⁴⁴

The thermal resistivity (R) was calculated as

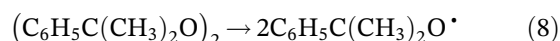
$$R = \frac{L}{k}. \quad (7)$$

Tensile tests were conducted on neat and xPP samples using an Instron universal testing machine (Model 5565, Instron, USA) with a 500 N load cell following ASTM D638 (type V) at a constant crosshead speed of 10 mm/min. The tensile modulus, tensile strength, and elongation at break were calculated by averaging results from at least four specimens, including standard deviations.

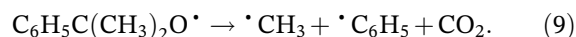
The surface hardness of the samples was measured using a model 306 L (Shore A) and model 307 L (Shore D) durometer (PTC Instruments, USA) according to ASTM D2240. Five measurements were taken for each sample (50 × 50 mm²) on both the internal and external surfaces to report averages and standard deviations.

3 | RESULTS AND DISCUSSIONS

During the rotational molding process, DCP serves as a crosslinking agent for PP. When the PP-DCP blends are heated above decomposition temperature (at 194°C), DCP decomposes into highly reactive radicals. These radicals play a dual role in the molding process: they initiate crosslinking and contribute to forming a cellular structure (in PP). The specific bond breaking during this process is the peroxide's oxygen–oxygen (O–O) link. This bond is relatively weak compared to other chemical bonds. As the temperature increases (194°C), the thermal energy becomes sufficient to break it. The breaking of this O–O bond can be represented as

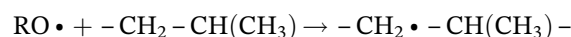


Because this is radical-based crosslinking, the radicals are available throughout the process for further reactions (production of by-products). These radicals can further break down to form methyl radicals (CH₃•) and phenyl radicals (C₆H₅•), releasing carbon dioxide (CO₂) as a by-product upon stabilization:

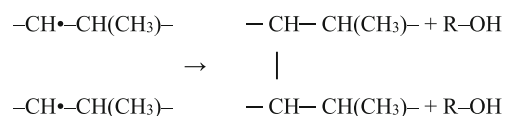


The generation of CO₂ at this step contributes to the potential formation of gas bubbles within the polymer matrix. Whereas the other highly reactive radicals, e.g., C₆H₅C(CH₃)₂O• (RO•), abstract hydrogen atoms from the PP backbone, which also leads to the formation of a cumyl alcohol (–OH) group as a byproduct⁴⁶ and leads to the formation of PP macroradicals that facilitate crosslinking such as follows:

I. Formation of crosslinks



II. Crosslinking of PP macro-radicals



This mechanism suggests that with increasing DCP concentration in PP, more radicals are generated, promoting crosslinking and producing gaseous by-products like CO₂. During rotational molding at 194°C, the low-pressure (no pressure) environment allows these gases to become trapped in the molten polymer, leading to bubble

formation. Rapid solidification during the cooling process further locks these bubbles in place, resulting in a more porous or foamed structure in crosslinked PP as DCP content increases.

Figure 3 presents the gel content (%) and crosslink density (mol/cm^3) of xPP with varying DCP content. Both the gel content and crosslink density increase with increasing DCP content. For neat PP (0% DCP), the gel content was found to be 0%. In contrast, for crosslinked polypropylene, the gel content increased by 11.3%, from 72.2% at 0.5% DCP to 80.4% at 2.5% DCP. Similar trends were observed for silane crosslinking systems to improve the foamability of PP.⁴² The increase in gel content is due to xylene interacting (at 140°C) with the polymer at the molecular level, swelling and dissolving the un-crosslinked fractions while preserving the crosslinked network. The polymer remains swollen below its melting point, ensuring selective removal of soluble chains, thus validating the method's effectiveness, consistent with prior studies.^{49–52} Additional experiments conducted at elevated temperatures (190°C using TCB, [Supplementary Materials](#)) confirm the reliability of these measurements. The gel content values obtained with TCB were similar to those with xylene, although stronger solvation power and harsher extraction conditions were used. Despite more variability (higher standard deviations), the gel content obtained via TCB supports the presence of a crosslinked network and validates the effectiveness of the crosslinking process. This confirms our findings and the suitability of the xylene-based method for gel content analysis under standard conditions. Furthermore, the crosslink density, which quantifies the number of crosslinks per unit volume of polymer, shows a significant increase (158%): from 1.2×10^{-3} to $3.1 \times 10^{-3} \text{ mol}/\text{cm}^3$ as the DCP content

increases from 0.5 to 2.5 phr. This increase underscores the formation of a more compact and interconnected polymer network. Higher DCP concentration introduces more peroxide molecules into the system, which subsequently decompose into more radicals. These radicals are highly reactive, thus efficiently initiating and propagating crosslinking reactions within the polymer matrix, leading to higher crosslink density.

Figure 4 presents the typical FTIR spectra for the neat PP and xPP samples. The typical bands for CH stretching are identified at 2950, 2916, and 2836 cm^{-1} , while the bands associated with CH_2 deformation and CH_3 symmetric deformation are observed at 1458 and 1378 cm^{-1} , respectively. The peaks for C–C backbone bending are observed at 1170 cm^{-1} . The interesting feature from the spectra is the presence of a signal at 3380 cm^{-1} related to the hydroxyl groups (OH) of crosslinking,^{43,46} and the peaks at 976 and 844 cm^{-1} indicating the isotactic nature of the polypropylene used.

Figure 5a–f presents SEM images at 75 $\times$ magnification, focusing on the bubble structures within the polymer matrix. No defects were observed for neat PP, as shown in Figure 5(A), which confirms the reliability of the processing parameters used during the rotational molding process. However, adding more DCP (increasing DCP content) leads to notable changes in the microstructure with varying bubble sizes and distributions, which are crucial to understanding the foaming behavior of the material.^{41,42} The decomposition of DCP (in rotomolding) releases gases, such as carbon dioxide (CO_2) generating bubbles in the molten polymer, along with other byproducts (acetophenone and cumyl alcohol). When the polymer begins to cool and solidify, the bubbles get trapped, resulting in a foam-like structure. In the context of

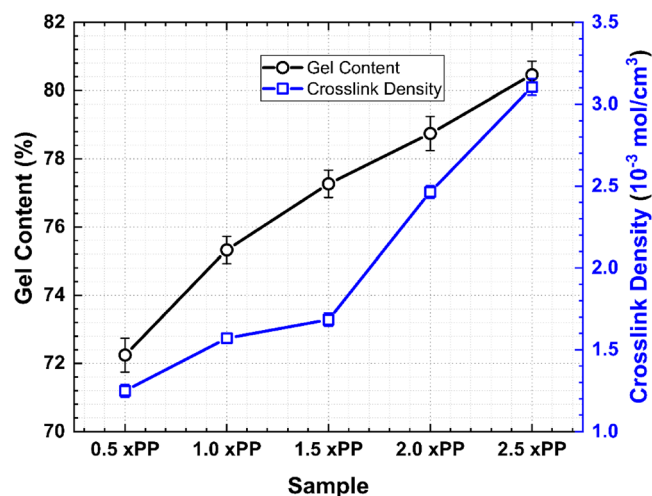


FIGURE 3 Gel content (%) and crosslink density (mol/cm^3) of xPP with different DCP content.

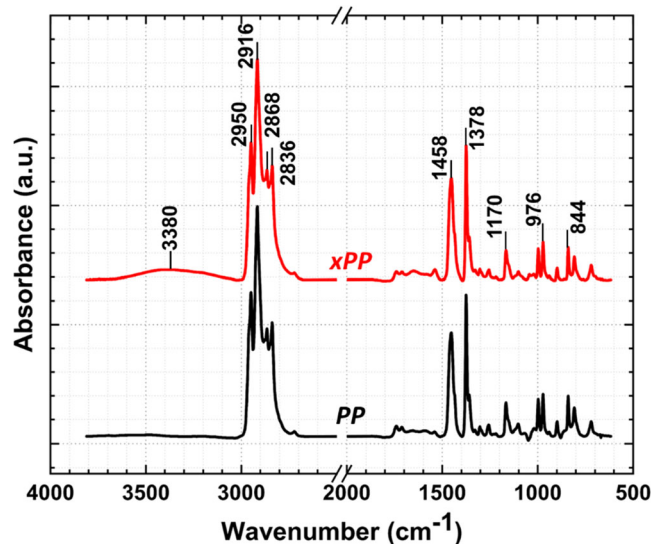


FIGURE 4 FTIR spectra of the neat PP and xPP (1.0 phr).

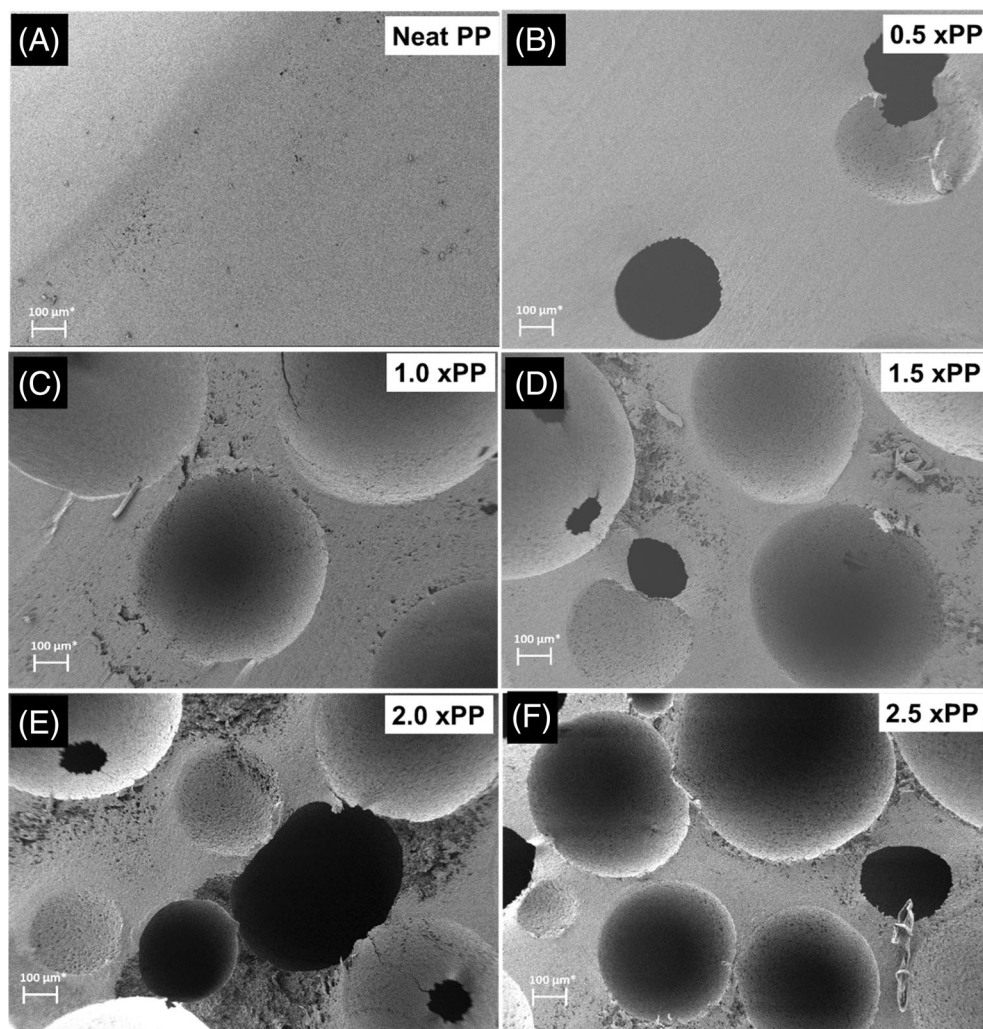


FIGURE 5 SEM micrographs (75 $\times$) of PP with different DCP content: (A) 0, (B) 0.5, (C) 1.0, (D) 1.5, (E) 2.0, and (F) 2.5 phr.

rotational molding, this trapping occurs effectively because the uniform heating and continuous rotation ensure that the molten polymer evenly coats the mold walls, allowing bubbles to expand and stabilize before solidification. Additionally, the low shear forces in rotational molding, combined with the absence of pressure in closed chambers (aluminum mold with a vent), further facilitate bubble formation, growth, and stabilization. Therefore, as the DCP content increases from 0.5 to 2.5 phr, a larger cell size (528–704 μm), as well as a higher cell density (1.5 to 9.6×10^3 cells/ cm^3) and expansion ratio (1–2.4) are obtained since more gas molecules are generated. A detailed observation related to these bubbles is presented in Figure 6a,b.

Following the analysis of bubble morphology (Figure 5), Figure 7 provides deeper insights into the structural integrity of the porous structure and cross-linked network formed with increasing DCP concentrations. The figure presents SEM images (a–e) at 1000 $\times$ magnification, focusing on the detailed architecture of the polymer structure (porous). With increasing DCP

content, the surface topography exhibits a notable increase in porosity. This is characterized by a higher number of both large (75 $\times$) and small pores, which are more clearly visible at higher magnifications (1000 $\times$). The formation of these pores is due to the generation and trapping of gaseous by-products, primarily CO_2 , during the crosslinking process (Equation 9). As more DCP is added, the amount of gas generated increases, leading to more extensive foaming within the polymer matrix. This results in a more heterogeneous structure with varying pore sizes, contributing to inhomogeneity in the bulk material and the cell walls of the bubbles. The presence of these pores, particularly the smaller ones, reduces the overall density of xPP (Figure 8), thereby improving its lightweight properties. While the intricate network of crosslinked polymer chains is challenging to visualize with SEM, the understanding of crosslink density and gel content provides a more comprehensive picture of the crosslinked structure (Figure 3).

The density and thickness of the neat PP and xPP with different DCP content are reported in Figure 8. As

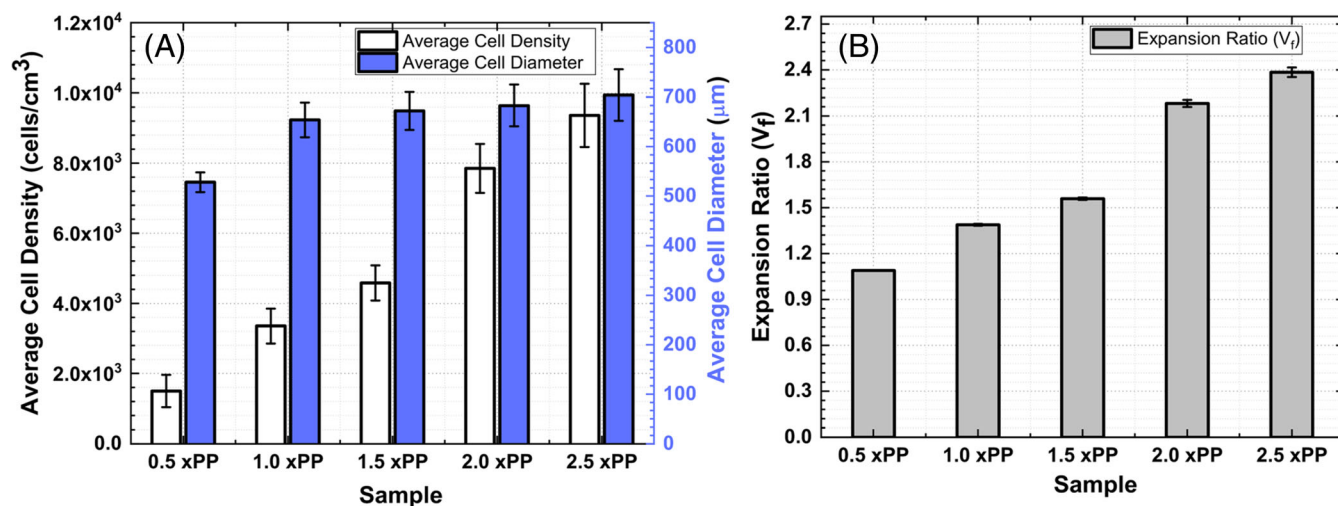
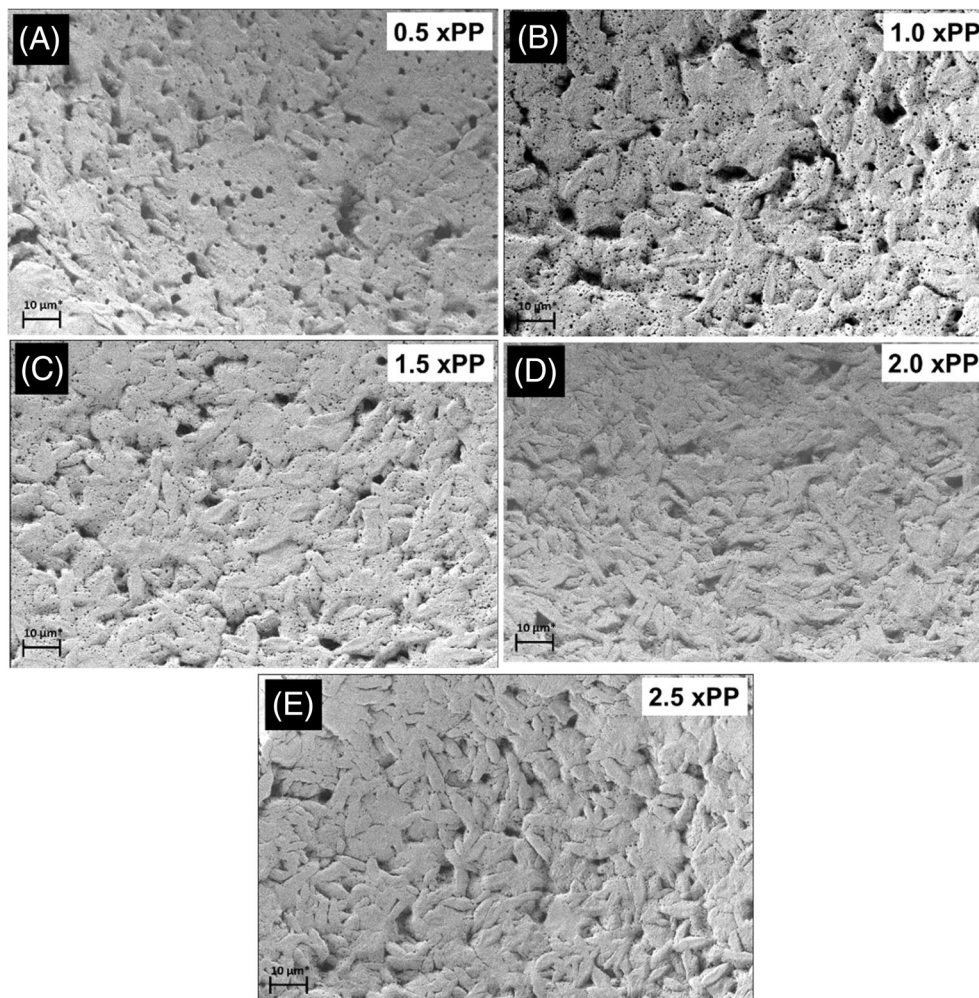


FIGURE 6 Microstructure analysis of the porous structure of xPP with various DCP content: (A) average cell density and diameter and (B) expansion ratio.

FIGURE 7 SEM micrographs (1000×) of PP with different DCP content: (A) 0.5, (B) 1.0, (C) 1.5, (D) 2.0, and (E) 2.5 phr.



the DCP content increases from 0.5 to 2.5 phr, a significant decrease in foam density is observed, dropping from 0.902 to 0.378 g/cm³. This trend is similar to that reported

by Hui et al.⁴² using vinyl trimethoxysilane (VTMS) as a crosslinking agent and azodicarbonamide (ACA) as a foaming agent to enhance PP foaming. Their study also

showed a reduction in foam density with increased cross-linker content, showing the effect of crosslinking on the final foam structure. This drastic reduction in density is attributed to the improved foaming behavior facilitated by the decomposition of DCP, which generates a higher volume of gas, mainly carbon dioxide. The increased gas release results in a more extensive formation of bubbles within the polymer matrix, leading to a lightweight and porous structure (Figure 8B). Concurrently, the thickness of the foam increases from 3.4 to 10.1 mm as a direct result of the greater expansion ratio (Figure 6B) associated with higher DCP content. Thicker parts indicate that the polymer is not only expanding more but also forming

a thinner cell wall structure supporting the foam integrity. As the expansion ratio increases, the gas bubbles within the foam structure grow larger. This greater expansion stretches the polymer matrix, causing the cell walls of these bubbles to become thinner because the polymer must cover a larger volume with the same amount of material, resulting in thinner walls between the bubbles. The combination of a low density and thin cell wall (more bubbles) foam structure confirms the effectiveness of DCP in promoting the formation of a stable and lightweight foam, which is critical for applications requiring excellent insulative properties and structural performance.

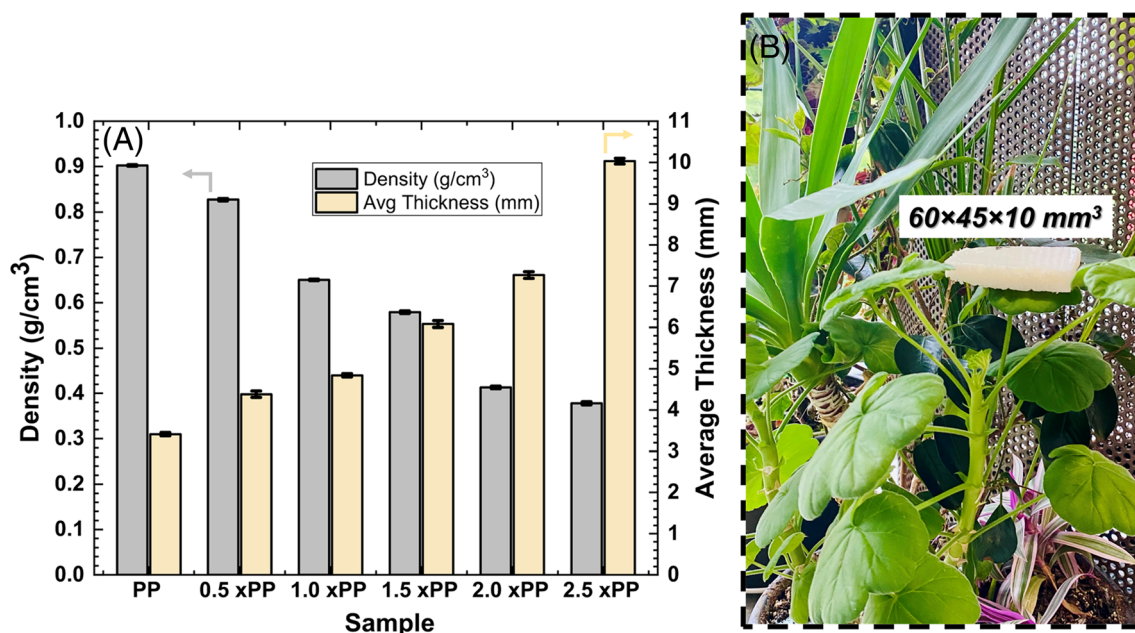


FIGURE 8 (A) Density and average thickness of the neat PP and xPP with various DCP contents and (B) 2.5 xPP sample on plant leaves showing its lightweight nature.

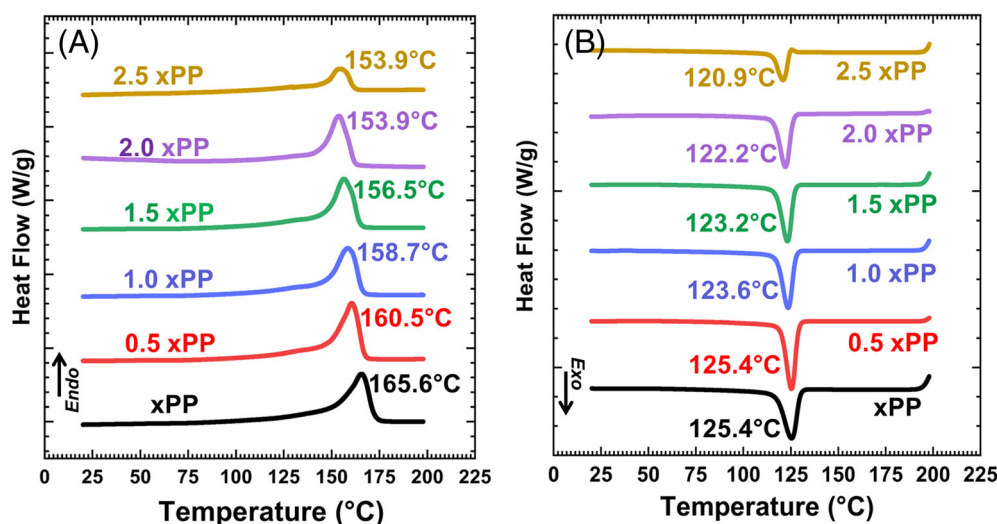


FIGURE 9 DSC thermograms of the neat PP and xPP with various DCP contents: (A) endothermic peaks (melting behavior) and (B) exothermic peaks (crystallization behavior).

Figure 9 presents the DSC thermograms of the neat PP and xPP with varying DCP content. From these thermograms, Table 1 summarizes the main properties obtained. The melting (T_m) and crystallization (T_c) temperatures, as well as the degree of crystallinity (X_c) all decrease with increasing DCP content. For example, T_m decreases from 165.6 to 153.9°C, T_c from 125.4 to 120.9°C, and X_c from 40.3% to 17.4% when the DCP content increases from 0.5 to 2.5 phr. These trends are related to the 3D network formed from crosslinking (with the addition of DCP), which disrupts the regular packing (crystalline regions) of the polymer molecules and reduces crystallinity. As a result, more of the polymer transitions into an amorphous or partially crystalline state. In polyolefins, the melting temperature is strongly and directly related to the crystalline regions.^{43,44,46}

Figure 10 shows the thermal conductivity (k) and resistivity (R) of the neat PP and xPP with various DCP contents over the range of temperatures studied (23–120°C). The thermal conductivity and resistivity of neat PP are 0.146 W/m·K and 0.023 (K/W), respectively, at 23°C, which is in agreement with the values reported in the literature.¹⁶ However, the addition of DCP leads to a decrease in thermal conductivity (62%) and high thermal

resistivity (691%). For example, the thermal conductivity of the xPP decreases from 0.146 to 0.055 W/m·K, while the thermal resistance increases from 0.023 to 0.182 K/W at 23°C when the DCP content increases to 2.5 phr. This trend is related to increasing DCP content promoting a more porous structure: a higher number of insulating gas bubbles inside the polymer matrix. These bubbles act as thermal insulators, reducing the overall heat transfer through the material along with a stronger 3D network restricting molecular motion.

Furthermore, as the temperature increases from 23 to 120°C, the polymer chain mobility improves, enhancing the energy transfer through molecular vibrations and collisions. This increased motion allows for better heat conduction, resulting in an increase in thermal conductivity (0.189 W/m·K) for neat PP at 120°C. However, a similar trend is observed when DCP is added; at 2.5 phr DCP, the thermal conductivity decreases (0.189–0.130 W/m·K at 120°C), while the thermal resistivity increases (0.003–0.010 K/W) at 120°C. This means that although increasing the temperature leads to an increase in thermal conductivity, adding more DCP leads to a reduction in thermal conductivity and an increase in thermal resistance, making the material suitable for higher insulative applications.

Table 2 presents the mechanical properties of the sample with varying DCP content. The tensile strength and modulus significantly decrease with increasing DCP content. The tensile strength drops by 97% (18.8 MPa for neat PP to 0.5 MPa for 2.5 xPP), while the tensile modulus decreases by 91% (282 MPa to 25.1 MPa). Similarly, both Shore A (17.2 points) and Shore D (34.2 points) hardness values show a notable decrease at 2.5 phr DCP content. These decreases are mainly attributed to the crosslinked structure (reduction in crystalline regions) and the dispersion of gas bubbles. As the DCP content

TABLE 1 Melting and crystallization properties of the neat PP and xPP with various DCP contents.

Sample	T_m (°C)	T_c (°C)	ΔH_m (J/g)	X_c (%)
PP	165.6	125.4	83.5	40.3
0.5 xPP	160.5	125.4	82.0	39.6
1.0 xPP	158.7	123.6	67.7	32.7
1.5 xPP	156.5	123.2	63.8	30.8
2.0 xPP	153.9	122.2	59.3	28.6
2.5 xPP	153.9	120.9	36.1	17.4

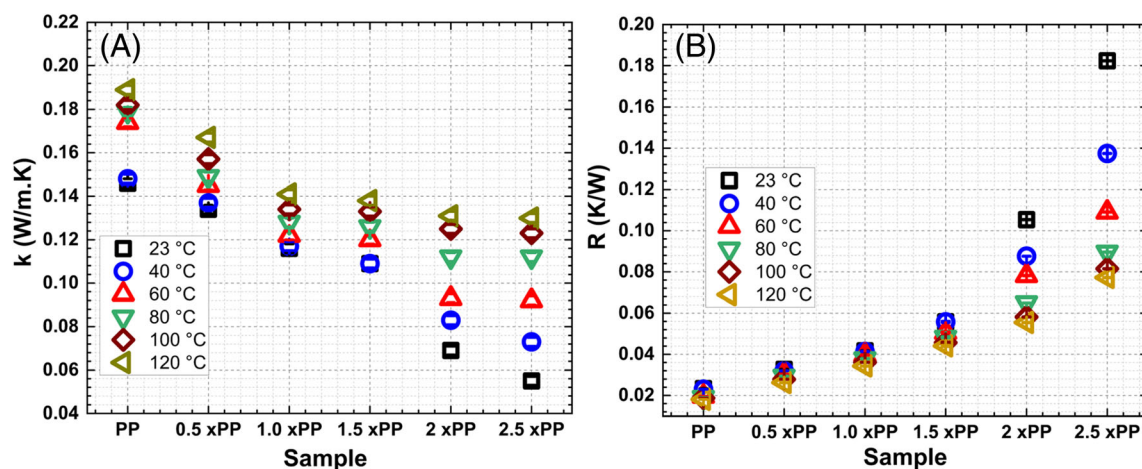


FIGURE 10 Temperature-dependent thermal conductivity and resistivity of the neat PP and xPP with various DCP contents.

TABLE 2 Mechanical properties of the neat PP and xPP with varying DCP content.

Sample	Tensile strength (MPa)	Tensile modulus (MPa)	Elongation at break (%)	Hardness (Shore A)	Hardness (Shore D)
PP	18.8 (0.7)	282 (47)	95 (16)	93 (1)	73 (1)
0.5 xPP	10.9 (0.8)	190 (31)	16.4 (0.4)	89 (1)	65 (1)
1.0 xPP	4.3 (0.5)	158 (24)	10.8 (0.4)	85 (1)	64 (1)
1.5 xPP	2.7 (0.6)	94 (32)	0.70 (0.01)	81 (1)	63 (1)
2.0 xPP	1.1 (0.2)	58 (1)	0.60 (0.05)	79 (1)	59 (1)
2.5 xPP	0.5 (0.1)	25 (1)	0.60 (0.05)	77 (1)	48 (1)

increases, more gas bubbles are introduced (more thin cell walls), creating a crosslinked porous structure (increasing cell size and density). The reduced thickness of the cell walls makes the foam more susceptible to deformation under mechanical stress.

Despite lower mechanical properties, this innovative approach offers a unique opportunity for further research by optimizing the processing conditions and exploring additional crosslinking agents or fillers (with and without nucleating agents) to improve the mechanical properties while retaining the foam's benefits.

4 | CONCLUSIONS

In this study, we successfully developed a lightweight, highly resistant crosslinked rotomolded polypropylene (xPP) with an innovative foam-like structure by incorporating varying dicumyl peroxide (DCP) content via rotational molding. The results showed significant improvements in thermal resistance, achieving an impressive 691% increase, while also significantly reducing the density to 0.378 g/cm³ at 2.5 phr DCP content. The study also revealed that increasing the DCP content up to 2.5 phr led to a corresponding increase in cell size (704 μm) and cell density (9.6×10^3 cells/cm³), confirming the successful formation of a porous structure contributing to the material's lightweight and insulative properties. However, this improvement in thermal performance and density reduction is associated with a trade-off as the mechanical properties, especially the tensile properties, were found to decrease with increasing DCP content. This highlights the challenge related to minimum lightweight and maximum thermal insulation coupled with mechanical strength/resistance for crosslinked polypropylene. Despite these opposite trends, the study provided an important benchmark for future research focusing on optimizing the formulations and processing conditions. Further investigations are needed to study and refine the crosslinking process, potentially improving the mechanical performance without compromising the material's lightweight and insulative advantages. In conclusion, this research

represents significant progress in the field of crosslinked porous rotomolded polypropylene. Its lightweight nature and high thermal insulation properties make it suitable for use in automotive interior components, such as door panels, headliners, and thermal insulation layers, as well as in construction for insulation boards or lightweight structural applications. Additionally, the crosslinked nature of the material improves its durability and dimensional stability, which are critical properties for applications in harsh environments (especially higher temperatures). Nevertheless, more work is needed to optimize the balance between crosslinking, foaming, and mechanical properties to meet the specific requirements of a broader range of applications. Modifications in composition, processing conditions, or the use of multi-layered structures could be explored to fine-tune these properties, ensuring that the molded products meet the performance of their intended applications. Finally, this study serves as a foundation for future research aimed at developing crosslinked foamed polypropylene materials with different structures.

ACKNOWLEDGMENTS

Financial support from the Association of Rotational Molders (ARM) was obtained via the Crawford Education and Development Foundation. PP powders from Rotoworx are also highly appreciated.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Denis Rodrigue  <https://orcid.org/0000-0002-3969-2847>

REFERENCES

- Wazeer A, Das A, Abeykoon C, Sinha A, Karmakar A. Composites for electric vehicles and automotive sector: a review. *Green Energy Intell Transp*. 2023;2:100043. doi:10.1016/j.geits.2022.100043

2. Kumar D, Alam M, Zou PW, Sanjayan JG, Memon RA. Comparative analysis of building insulation material properties and performance. *Renew Sust Energ Rev*. 2020;131:110038. doi:10.1016/j.rser.2020.110038
3. Borghesi M, Ghassemi M. Insulation materials and systems for more- and all-electric aircraft: a review identifying challenges and future research needs. *IEEE Trans Transp Electr*. 2021;7:1930-1953. doi:10.1109/TTE.2021.3050269
4. Cai Z, Al-Farouque MA, Kiziltas A, Mielewski D, Naebe M. Sustainable lightweight insulation materials from textile-based waste for the automobile industry. *Materials*. 2021;14:1241. doi:10.3390/ma14051241
5. Zheng L, Zuo Y, Li X, Wu Y. Biomimetic swallow Nest structure: a lightweight and high-strength thermal insulation material. *ACS Nano*. 2022;16:8116-8127. doi:10.1021/acsnano.2c01451
6. Wu D, Lin L, Ren H. Thermal/vibration joint experimental investigation on lightweight ceramic insulating material for hypersonic vehicles in extremely high-temperature environment up to 1500°C. *Ceram Int*. 2020;46:14439-14447. doi:10.1016/j.ceramint.2020.02.241
7. Lizcano M, Williams TS, Shin EE, Santiago D, Nguyen B. Aerospace environmental challenges for electrical insulation and recent developments for electrified aircraft. *Mater Basel*. 2022;15(22):8121. doi:10.3390/ma15228121
8. Nagaraju SB, Priya HC, Girijappa YT, Puttegowda M. *Light-weight and Sustainable Materials for Aerospace Applications*. Woodhead Publishing; 2023:157-178. doi:10.1016/B978-0-323-95189-0.00007-X
9. Jelčić RM, Skejić D, Kralj A, Ščapeć T, Milovanović B. Development of lightweight steel framed construction systems for nearly-zero energy buildings. *Buildings*. 2022;12(7):929. doi:10.3390/buildings12070929
10. Xu Y, Yan C, Du C, et al. High-strength, thermal-insulating, fire-safe bio-based organic lightweight aerogel based on 3D network construction of natural tubular fibers. *Compos Part B Eng*. 2023;261:110809. doi:10.1016/j.compositesb.2023.110809
11. Laxmi V, Thakre C, Vijay R. Evaluation of noise barriers based on geometries and materials: a review. *Environ Sci Pollut Res*. 2022;29:1729-1745. doi:10.1007/s11356-021-16944-2
12. Zhao W, Zhao HB, Cheng JB, Li W, Zhang J, Wang YZ. A green, durable and effective flame-retardant coating for expandable polystyrene foams. *Chem Eng J*. 2022;440:135807. doi:10.1016/j.cej.2022.135807
13. Wang G, Zhao J, Mark LH, et al. Ultra-tough and super thermal-insulation nanocellular PMMA/TPU. *Chem Eng J*. 2017;325:632-646. doi:10.1016/j.cej.2017.05.116
14. Dou Y, Rodrigue D. Rotomolding of foamed and unfoamed GTR-LLDPE blends: mechanical, morphological and physical properties. *Cell Polym*. 2018;37:55-68. doi:10.1177/026248931803700201
15. Wong S, Lee JS, Naguib HE, Park CB. Effect of processing parameters on the mechanical properties of injection molded thermoplastic polyolefin (TPO) cellular foams. *Macromol Mater Eng*. 2008;293:605-613. doi:10.1002/mame.200700362
16. Dou Y, Rodrigue D. Morphological, thermal and mechanical properties of polypropylene foams via rotational molding. *Cell Polym*. 2021;40:198-211. doi:10.1177/02624893211018825
17. Zhong W, Hu D, Chen Y, Zhao L. Microwave-assisted physical foaming of polymers: a review. *Polym Rev*. 2024;64(4):1136-1175. doi:10.1080/15583724.2024.2372492
18. Kuhnigk J, Standau T, Dörr D, Brütting C, Altstädt V, Ruckdäschel H. Progress in the development of bead foams – a review. *J Cell Plast*. 2022;58:707-735. doi:10.1177/0021955X221087603
19. Rostami M, Azdast T, Hasanazadeh R, Moradian M. A study on fabrication of nanocomposite polyethylene foam through extrusion foaming procedure. *Cell Polym*. 2021;40:231-243. doi:10.1177/02624893211040949
20. Jiang J, Li Z, Yang H, Wang X, Li Q, Turng LS. Microcellular injection molding of polymers: a review of process know-how, emerging technologies, and future directions. *Curr Opin Chem Eng*. 2021;33:100694. doi:10.1016/j.coche.2021.100694
21. Lee JK, Yao SX, Li G, Jun MBG, Lee PC. Measurement methods for solubility and diffusivity of gases and supercritical fluids in polymers and its applications. *Polym Rev*. 2017;57:695-747. doi:10.1080/15583724.2017.1329209
22. Lee J, Yao SX, Lee JK, Lee PC. A new gas/supercritical fluid (SCF) diffusivity measurement method for CO₂ saturated polymer systems using optical transmission property. *TechConnect World Innovation Conference and Expo Washington, DC*. Vol 5. TechConnect; 2017:14-17.
23. Yao S, Lee P. A new gas/supercritical fluid (SCF) diffusivity measurement method for CO₂ saturated polymer systems using a dielectric property. *Express Polym Lett*. 2017;11:664-672. doi:10.3144/expresspolymlett.2017.64
24. Shubhra QH, Alam A, Quaiyyum MA. Mechanical properties of polypropylene composites: a review. *J Thermoplast Compos Mater*. 2011;26:362-391. doi:10.1177/0892705711428659
25. Busico V, Cipullo R. Microstructure of polypropylene. *Prog Polym Sci*. 2001;26:443-533. doi:10.1016/S0079-6700(00)00046-0
26. Tsiptsias C, Leontiadis K, Tzimpilis E, Tsivintzelis I. Polypropylene nanocomposite fibers: a review of current trends and new developments. *J Plast Film Sheet*. 2020;37:283-311. doi:10.1177/8756087920972146
27. Wang K, Wu F, Zhai W, Zheng W. Effect of polytetrafluoroethylene on the foaming behaviors of linear polypropylene in continuous extrusion. *J Appl Polym Sci*. 2013;129:2253-2260. doi:10.1002/app.38959
28. Wong A, Guo Y, Park CB. Fundamental mechanisms of cell nucleation in polypropylene foaming with supercritical carbon dioxide effects of extensional stresses and crystals. *J Supercrit Fluids*. 2013;79:142-151. doi:10.1016/j.supflu.2013.02.013
29. Fereidoon A, Memarian S, Albooyeh A, Tarahomi S. Influence of mesoporous silica and hydroxyapatite nanoparticles on the mechanical and morphological properties of polypropylene. *Mater Des*. 2014;57:201-210. doi:10.1016/j.matdes.2013.12.046
30. Colton JS. The nucleation of microcellular foams in semi crystalline thermoplastics*. *Mater Manuf Process*. 1989;4:253-262. doi:10.1080/10426918908956288
31. Zhao J, Zhao Q, Wang C, Guo B, Park CB, Wang G. High thermal insulation and compressive strength polypropylene foams fabricated by high-pressure foam injection molding and mold opening of nano-fibrillar composites. *Mater Des*. 2017;131:1-11. doi:10.1016/j.matdes.2017.05.093
32. Rizvi A, Chu RM, Lee JH, Park CB. Superhydrophobic and oleophilic open-cell foams from fibrillar blends of

- polypropylene and polytetrafluoroethylene. *ACS Appl Mater Interfaces*. 2014;6:21131-21140. doi:[10.1021/am506006v](https://doi.org/10.1021/am506006v)
33. Rizvi A, Tabatabaei A, Barzegari MR, Mahmood SH, Park CB. In situ fibrillation of CO₂-Philic polymers: sustainable route to polymer foams in a continuous process. *Polymer (Guildf)*. 2013; 54:4645-4652. doi:[10.1016/j.polymer.2013.06.023](https://doi.org/10.1016/j.polymer.2013.06.023)
 34. Yang C, Zhang Q, Zhang W, et al. High thermal insulation and compressive strength polypropylene microcellular foams with honeycomb structure. *Polym Degrad Stab*. 2021;183:109406. doi:[10.1016/j.polymdegradstab.2020.109406](https://doi.org/10.1016/j.polymdegradstab.2020.109406)
 35. Nam PH, Maiti P, Okamoto M, et al. Foam processing and cellular structure of polypropylene/clay nanocomposites. *Polym Eng Sci*. 2002;42:1907-1918. doi:[10.1002/pen.11083](https://doi.org/10.1002/pen.11083)
 36. Naguib HE, Park CB, Reichelt N. Fundamental foaming mechanisms governing the volume expansion of extruded polypropylene foams. *J Appl Polym Sci*. 2004;91:2661-2668. doi:[10.1002/app.13448](https://doi.org/10.1002/app.13448)
 37. Mohebbi A, Mighri F, Aji A, Rodrigue D. Effect of processing conditions on the cellular morphology of polypropylene foamed films for piezoelectric applications. *Cell Polym*. 2017;36: 13-34. doi:[10.1177/026248931703600102](https://doi.org/10.1177/026248931703600102)
 38. Stumpf M, Spörrer A, Schmidt HW, Altstädt V. Influence of supramolecular additives on foam morphology of injection-molded i-PP. *J Cell Plast*. 2011;47:519-534. doi:[10.1177/0021955X11408769](https://doi.org/10.1177/0021955X11408769)
 39. Mörl M, Steinlein C, Kreger K, Schmidt HW, Altstädt V. Improved compression properties of polypropylene extrusion foams by supramolecular additives. *J Cell Plast*. 2017;54:483-498. doi:[10.1177/0021955X17695096](https://doi.org/10.1177/0021955X17695096)
 40. Soares FA, Nachtigall SB. Effect of chemical and physical foaming additives on the properties of PP/wood flour composites. *Polym Test*. 2013;32:640-646. doi:[10.1016/j.polymertesting.2013.02.009](https://doi.org/10.1016/j.polymertesting.2013.02.009)
 41. Zhang ZX, Wang C, Wang S, Wen S, Phule AD. A lightweight, thermal insulation and excellent weatherability foam cross-linked by electron beam irradiation. *Radiat Phys Chem*. 2020; 173:108890. doi:[10.1016/j.radphyschem.2020.108890](https://doi.org/10.1016/j.radphyschem.2020.108890)
 42. Liu H, Chuai C, Iqbal M, et al. Improving foam ability of polypropylene by crosslinking. *J Appl Polym Sci*. 2011;122:973-980. doi:[10.1002/app.33227](https://doi.org/10.1002/app.33227)
 43. Ahmad H, Rostam E, Rodrigue D. The effect of chemical crosslinking on the properties of rotomolded high density polyethylene. *J Appl Polym Sci*. 2024;141:e54744. doi:[10.1002/app.54744](https://doi.org/10.1002/app.54744)
 44. Ahmad H, Rodrigue D. High-performance wood-reinforced crosslinked high-density polyethylene composites. *Polym Eng Sci*. 2024;64:2459-2475. doi:[10.1002/pen.26702](https://doi.org/10.1002/pen.26702)
 45. Ahmad H, Rodrigue D. Upcycling of recycled polyethylene for rotomolding applications via dicumyl peroxide crosslinking. *J Appl Polym Sci*. 2024;141:e56236. doi:[10.1002/app.56236](https://doi.org/10.1002/app.56236)
 46. Ahmad H, Rodrigue D. Crosslinked polyethylene: a review on the crosslinking techniques, manufacturing methods, applications, and recycling. *Polym Eng Sci*. 2022;62:2376-2401. doi:[10.1002/pen.26049](https://doi.org/10.1002/pen.26049)
 47. Flory PJ, Rehner JJ. Statistical mechanics of cross-linked polymer networks II. Swelling. *J Chem Phys*. 1943;11:521-526. doi:[10.1063/1.1723792](https://doi.org/10.1063/1.1723792)
 48. Gee DR, Melia TP. Thermal properties of melt and solution crystallized isotactic polypropylene. *Macromol Chem*. 1970;132: 195-201. doi:[10.1002/macp.1970.021320117](https://doi.org/10.1002/macp.1970.021320117)
 49. Gravgaard DP, Henriksen ML, Hinge M. Dissolution recycling for recovery of polypropylene and glass Fibres. *J Mater Cycles Waste Manag*. 2024;26(2):961-969. doi:[10.1007/s10163-023-01873-8](https://doi.org/10.1007/s10163-023-01873-8)
 50. Wang Z, Wu X, Gui Z, Hu Y, Fan W. Thermal and crystallization behavior of silane-crosslinked polypropylene. *Polym Int*. 2005;54(2):442-447. doi:[10.1002/pi.1715](https://doi.org/10.1002/pi.1715)
 51. Yoshiga A, Otaguro H, Parra DF, Lima LFCP, Lugao AB. Controlled degradation and crosslinking of polypropylene induced by gamma radiation and acetylene. *Polym Bull*. 2009;63(3):397-409. doi:[10.1007/s00289-009-0102-7](https://doi.org/10.1007/s00289-009-0102-7)
 52. Yu Q, Zhu S. Peroxide crosslinking of isotactic and syndiotactic polypropylene. *Polymer (Guildf)*. 1999;40(11):2961-2968. doi:[10.1016/S0032-3861\(98\)00519-9](https://doi.org/10.1016/S0032-3861(98)00519-9)

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Ahmad H, Rodrigue D. Development of lightweight and thermally insulative crosslinked polypropylene via rotomolding. *Polym Eng Sci*. 2025;65(6):3059-3072. doi:[10.1002/pen.27198](https://doi.org/10.1002/pen.27198)